Appendix

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A. Proofs

A.1. Preliminaries

Proposition 1. *Given $G = (A,R)$. We have $stb(G) \subseteq prf(G) \subseteq cmp(G)$.*

Proof. It can be found in Dung's paper[1]. □

Proposition 2 (Preservation of extensions under structural forgetting)**.** *Given $F = (A,D,Sub)$ and $\Gamma(F) = (A,D_\Gamma)$ its forgetful projection. For every σ_{SAF} and its corresponding classical semantics σ_{AAF}, it holds that $\sigma_{\mathrm{SAF}}(F) = \sigma_{\mathrm{AAF}}(\Gamma(F))$.*

Proof. It can be found in Liao's paper[**?**]. □

A.2. Probabilistic Subargument Argumentation Framework

Lemma 1. *Let $PrF = (F,P)$, where $F = (A,D,Sub)$, and let $F_{\downarrow W} = (A_{\downarrow W}, D_{\downarrow W}, Sub_{\downarrow W})$ be a possible world of PrF. For every argument $a \in A$, $a \in W$ iff $\mathrm{Sub_b}(a) \subseteq \mathrm{Base}(F_{\downarrow W})$.*

Proof. Let $PrF = (F,P)$, where $F = (A,D,Sub)$, and let $F_{\downarrow W} = (A_{\downarrow W}, D_{\downarrow W}, Sub_{\downarrow W})$ be a possible world of PrF. Since $A_{\downarrow W} = W$, it suffices to show that for every $a \in A$, $a \in A_{\downarrow W}$ iff $\mathrm{Sub_b}(a) \subseteq \mathrm{Base}(F_{\downarrow W})$.

($\Rightarrow$) Assume that $a \in A_{\downarrow W}$. Let $b \in \mathrm{Sub_b}(a)$. Then $b \in \mathrm{Base}(F)$ and $(b,a) \in Sub^*$. Since $F_{\downarrow W}$ is a possible world and $a \in A_{\downarrow W}$, repeated application of subargument closure yields $b \in A_{\downarrow W}$. It remains to show that $b \in \mathrm{Base}(F_{\downarrow W})$. If not, then b has some direct subargument in $F_{\downarrow W}$, say $c \in A_{\downarrow W}$ with $(c,b) \in Sub_{\downarrow W}$. Since $Sub_{\downarrow W} \subseteq Sub$, this gives $(c,b) \in Sub$, contradicting $b \in \mathrm{Base}(F)$. Hence $b \in \mathrm{Base}(F_{\downarrow W})$. Therefore $\mathrm{Sub_b}(a) \subseteq \mathrm{Base}(F_{\downarrow W})$.

($\Leftarrow$) Assume that $\mathrm{Sub_b}(a) \subseteq \mathrm{Base}(F_{\downarrow W})$. Then every basic subargument of a belongs to $A_{\downarrow W}$. We prove $a \in A_{\downarrow W}$ by induction on the depth of a in the acyclic subargument relation. If a is basic, then $\mathrm{Sub_b}(a) = \{a\}$, so $a \in \mathrm{Base}(F_{\downarrow W}) \subseteq A_{\downarrow W}$. Now suppose a is non-basic, and assume the claim holds for every proper subargument of a. Let $d \in Sub_d(a)$. Every basic subargument of d is also a basic subargument of a, so $\mathrm{Sub_b}(d) \subseteq \mathrm{Sub_b}(a) \subseteq \mathrm{Base}(F_{\downarrow W})$. By the induction hypothesis, $d \in A_{\downarrow W}$. Since this holds for every $d \in Sub_d(a)$, we have $Sub_d(a) \subseteq A_{\downarrow W}$. As a is non-basic, $Sub_d(a) \neq \emptyset$, and by structural completion of possible worlds it follows that $a \in A_{\downarrow W}$.

Thus $a \in W$ iff $\mathrm{Sub_b}(a) \subseteq \mathrm{Base}(F_{\downarrow W})$. □

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Theorem 1 (Existence and Uniqueness of the Generated Possible World). *Let $PrF = (F,P)$, where $F = (A,D,Sub)$, and let $B \subseteq \mathrm{Base}(F)$ be a set of basic arguments in F. Then there exists a unique possible world $F_B = (A_B, D_B, Sub_B)$ of PrF such that $\mathrm{Base}(F_B) = B$.*

Proof. For the given set $B \subseteq \mathrm{Base}(F)$, define $A_B = \{a \in A \mid \mathrm{Sub_b}(a) \subseteq B\}$, $D_B = D \cap (A_B \times A_B)$, and $Sub_B = Sub \cap (A_B \times A_B)$. We show that $F_B = (A_B, D_B, Sub_B)$ is the unique possible world of PrF such that $\mathrm{Base}(F_B) = B$.

First, F_B is a subframework of F by construction. We verify the two conditions for possible worlds. For subargument closure, let $a \in A_B$ and let $d \in Sub_d(a)$. Since every basic subargument of d is also a basic subargument of a, we have $\mathrm{Sub_b}(d) \subseteq \mathrm{Sub_b}(a) \subseteq B$, so $d \in A_B$. Hence $Sub_d(a) \subseteq A_B$. For structural completion, let $a \in A$ satisfy $Sub_d(a) \subseteq A_B$ and $Sub_d(a) \neq \emptyset$. We show that $a \in A_B$. Let $b \in \mathrm{Sub_b}(a)$. Since b is a basic subargument of a, there is a chain from b to a in Sub whose last step goes through some direct subargument $d \in Sub_d(a)$. Thus $b \in \mathrm{Sub_b}(d)$. As $d \in A_B$, we have $\mathrm{Sub_b}(d) \subseteq B$, and therefore $b \in B$. Since this holds for every $b \in \mathrm{Sub_b}(a)$, we get $\mathrm{Sub_b}(a) \subseteq B$, i.e. $a \in A_B$. Hence F_B is a possible world.

Next, we show that $\mathrm{Base}(F_B) = B$. If $b \in B$, then $b \in \mathrm{Base}(F)$, so $\mathrm{Sub_b}(b) = \{b\} \subseteq B$, whence $b \in A_B$. Since b has no direct subarguments in F, it also has none in F_B, so $b \in \mathrm{Base}(F_B)$. Thus $B \subseteq \mathrm{Base}(F_B)$. Conversely, let $a \in \mathrm{Base}(F_B)$. If $a \notin \mathrm{Base}(F)$, then there exists $d \in Sub_d(a)$. Since $a \in A_B$, we have $\mathrm{Sub_b}(a) \subseteq B$, and therefore $\mathrm{Sub_b}(d) \subseteq \mathrm{Sub_b}(a) \subseteq B$, so $d \in A_B$. Hence $(d,a) \in Sub_B$, contradicting $a \in \mathrm{Base}(F_B)$. Therefore $a \in \mathrm{Base}(F)$. Since $a \in A_B$, we have $\mathrm{Sub_b}(a) = \{a\} \subseteq B$, so $a \in B$. Thus $\mathrm{Base}(F_B) \subseteq B$, and therefore $\mathrm{Base}(F_B) = B$.

Finally, uniqueness follows from Lemma 1. Let $F' = (A', D', Sub')$ be any possible world of PrF such that $\mathrm{Base}(F') = B$. For every $a \in A$, Lemma 1 gives $a \in A'$ iff $\mathrm{Sub_b}(a) \subseteq \mathrm{Base}(F') = B$, iff $a \in A_B$. Hence $A' = A_B$. Since a possible world is uniquely determined by its argument set, we obtain $D' = D \cap (A_B \times A_B) = D_B$ and $Sub' = Sub \cap (A_B \times A_B) = Sub_B$. Therefore $F' = F_B$.

So there exists a unique possible world $F_B = (A_B, D_B, Sub_B)$ such that $\mathrm{Base}(F_B) = B$. □

Proposition 3 (Normalization). *Let $PrF = (F,P)$, where $F = (A,D,Sub)$, and let Ω_{PrF} be the set of all possible worlds of PrF. The world probability assignment* $\Pr$ *is a probability distribution on Ω_{PrF}, in particular, $\sum_{F_B \in \Omega_{PrF}} \Pr(F_B) = 1$.*

Proof. Let $\mathrm{Base}(F) = \{b_1, \ldots, b_n\}$. By Theorem 1, for every subset $B \subseteq \mathrm{Base}(F)$ there exists a unique possible world $F_B \in \Omega_{PrF}$ such that $\mathrm{Base}(F_B) = B$, and every possible world of PrF is of this form. Hence

$$\sum_{F_B \in \Omega_{PrF}} \Pr(F_B) = \sum_{B \subseteq \mathrm{Base}(F)} \Pr(F_B).$$

By Definition 5, for every $B \subseteq \mathrm{Base}(F)$,

$$\Pr(F_B) = \prod_{b \in B} P(b) \prod_{b \in \mathrm{Base}(F) \setminus B} (1 - P(b)).$$

Therefore

$$\sum_{F_B\in\Omega_{PrF}} \Pr(F_B) = \sum_{B\subseteq \mathrm{Base}(F)} \left(\prod_{b\in B} P(b) \prod_{b\in \mathrm{Base}(F)\setminus B} (1-P(b))\right) = \prod_{b\in \mathrm{Base}(F)} (P(b)+(1-P(b))) = 1.$$

Thus Pr is a probability distribution on Ω_{PrF}. □

Proposition 4 (Consistency)**.** *Let $PrF = (F,P)$, where $F = (A,D,Sub)$, and let Ω_{PrF} be the set of all possible worlds of PrF. Then, for every argument $a \in A$, $P(a) = \sum_{F_B\in\Omega_{PrF},a\in A_B} \Pr(F_B)$.*

Proof. Fix $a \in A$. By Theorem 1, every possible world in Ω_{PrF} is uniquely of the form F_B for some $B \subseteq \mathrm{Base}(F)$. By Lemma 1, for every such F_B we have $a \in A_B$ iff $\mathrm{Sub_b}(a) \subseteq \mathrm{Base}(F_B) = B$. Hence

$$\sum_{F_B\in\Omega_{PrF},a\in A_B} \Pr(F_B) = \sum_{B\subseteq \mathrm{Base}(F),\mathrm{Sub_b}(a)\subseteq B} \left(\prod_{b\in B} P(b) \prod_{b\in \mathrm{Base}(F)\setminus B} (1-P(b))\right).$$

Write $S = \mathrm{Sub_b}(a)$. Then

$$\sum_{B\subseteq \mathrm{Base}(F),S\subseteq B} \left(\prod_{b\in B} P(b) \prod_{b\in \mathrm{Base}(F)\setminus B} (1-P(b))\right) = \prod_{b\in S} P(b) \sum_{T\subseteq \mathrm{Base}(F)\setminus S} \left(\prod_{b\in T} P(b) \prod_{b\in (\mathrm{Base}(F)\setminus S)\setminus T} (1-P(b))\right).$$

It is similar in Proposition 3, so

$$\sum_{F_B\in\Omega_{PrF},a\in A_B} \Pr(F_B) = \prod_{b\in S} P(b) \prod_{b\in \mathrm{Base}(F)\setminus S} (P(b)+(1-P(b))) = \prod_{b\in \mathrm{Sub_b}(a)} P(b).$$

By Definition 3 of structure-compliant probability assignment, $\prod_{b\in\mathrm{Sub_b}(a)} P(b) = P(a)$. Therefore

$$P(a) = \sum_{F_B\in\Omega_{PrF},a\in A_B} \Pr(F_B).$$

□

A.3. Properties

Postulate 1 (Semantic Preservation)**.** *$PrF \in \mathscr{U}_{PSAF}$ satisfies* semantic preservation *iff for every $E \subseteq A$, it holds that $\Pr^{F,P}_{STB}(E) \le \Pr^{F,P}_{PRF}(E) \le \Pr^{F,P}_{CMP}(E)$.*

Postulate 2 (Acceptance Bounds)**.** *$PrF \in \mathscr{U}_{PSAF}$ satisfies* acceptance bounds *iff for every σ_{SAF} and every $a \in A$, it holds that $0 \le \Pr^{F,P}_{sk,\sigma_{\mathrm{SAF}}}(a) \le \Pr^{F,P}_{cr,\sigma_{\mathrm{SAF}}}(a) \le P(a)$.*

Postulate 3 (Conflict Handling)**.** *$PrF \in \mathscr{U}_{PSAF}$ satisfies* conflict handling *iff for every σ_{SAF} and every $a,b \in A$, if a and b are in conflict, then $\Pr^{F,P}_{sk,\sigma_{\mathrm{SAF}}}(a) + \Pr^{F,P}_{sk,\sigma_{\mathrm{SAF}}}(b) \le 1$.*

Theorem 2. *Every $PrF \in \mathscr{U}_{PSAF}$ satisfies Postulates 1–3.*

Proof. Let $PrF = (F,P)$, where $F = (A,D,Sub)$, and let Ω_{PrF} be the set of all possible worlds of PrF.

For Postulate 1, fix $E \subseteq A$. For every $F_B \in \Omega_{PrF}$, the inclusion relation among structure-sensitive semantics gives $STB(F_B) \subseteq PRF(F_B) \subseteq CMP(F_B)$ by Proposition 1 and Proposition 2. Hence $\{F_B \in \Omega_{PrF} \mid E \in STB(F_B)\} \subseteq \{F_B \in \Omega_{PrF} \mid E \in PRF(F_B)\} \subseteq \{F_B \in \Omega_{PrF} \mid E \in CMP(F_B)\}$. Summing the world probabilities over these nested sets yields $\Pr^{F,P}_{STB}(E) \leq \Pr^{F,P}_{PRF}(E) \leq \Pr^{F,P}_{CMP}(E)$.

For Postulate 2, fix σ_{SAF} and $a \in A$. By definition 7, $\Pr^{F,P}_{sk,\sigma_{\text{SAF}}}(a)$ and $\Pr^{F,P}_{cr,\sigma_{\text{SAF}}}(a)$ are sums of non-negative world probabilities, so $0 \leq \Pr^{F,P}_{sk,\sigma_{\text{SAF}}}(a)$ and $0 \leq \Pr^{F,P}_{cr,\sigma_{\text{SAF}}}(a)$. Moreover, every world in which a is skeptically accepted is also a world in which a is credulously accepted, so $\Pr^{F,P}_{sk,\sigma_{\text{SAF}}}(a) \leq \Pr^{F,P}_{cr,\sigma_{\text{SAF}}}(a)$. Finally, if a is credulously accepted in F_B, then necessarily $a \in A_B$, since only occurring arguments can belong to a σ_{SAF}-extension. Therefore $\Pr^{F,P}_{cr,\sigma_{\text{SAF}}}(a) \leq \sum_{F_B \in \Omega_{PrF}, a \in A_B} \Pr(F_B) = P(a)$, where the equality is Proposition 4. Hence $0 \leq \Pr^{F,P}_{sk,\sigma_{\text{SAF}}}(a) \leq \Pr^{F,P}_{cr,\sigma_{\text{SAF}}}(a) \leq P(a)$.

For Postulate 3, fix σ_{SAF} and let $a,b \in A$ be in conflict. If some world F_B made both a and b skeptically accepted, then every σ_{SAF}-extension of F_B would contain both a and b. But every σ_{SAF}-extension is conflict-free, so no such extension can contain two conflicting arguments. Hence the sets $\{F_B \in \Omega_{PrF} \mid F_B \mid\!\sim^{sk}_{\sigma_{\text{SAF}}} a\}$ and $\{F_B \in \Omega_{PrF} \mid F_B \mid\!\sim^{sk}_{\sigma_{\text{SAF}}} b\}$ are disjoint. Therefore $\Pr^{F,P}_{sk,\sigma_{\text{SAF}}}(a) + \Pr^{F,P}_{sk,\sigma_{\text{SAF}}}(b) \leq \sum_{F_B \in \Omega_{PrF}} \Pr(F_B) = 1$.

Thus every $PrF \in \mathscr{U}_{PSAF}$ satisfies Postulates 1–3. □

For each comparative postulate above, its PAAF counterpart is obtained by replacing PSAF specific notation to corresponding noation in PAAF.

Theorem 3. *For every $PrG \in \mathscr{U}_{PAAF}$, PrG satisfies the counterparts of Postulates 1–3.*

Proof. Let $PrG = (G,P)$, where $G = (A,R)$, and let Ω_{PrG} be the set of all possible worlds of PrG.

For the counterpart of Postulate 1 and Postulate 2, the proof can be found in Hunter's paper[2]. And the proof strategy is similar to the Theorem 2.

For the counterpart of Postulate 3, fix σ_{AAF} and let $a,b \in A$ be in conflict, i.e. $(a,b) \in R$ or $(b,a) \in R$. If some world $G_{\downarrow W}$ made both a and b skeptically accepted, then every σ_{AAF}-extension of $G_{\downarrow W}$ would contain both a and b. But every σ_{AAF}-extension is conflict-free, so no such extension can contain two conflicting arguments. Hence the sets $\{G_{\downarrow W} \in \Omega_{PrG} \mid G_{\downarrow W} \mid\!\sim^{sk}_{\sigma_{\text{AAF}}} a\}$ and $\{G_{\downarrow W} \in \Omega_{PrG} \mid G_{\downarrow W} \mid\!\sim^{sk}_{\sigma_{\text{AAF}}} b\}$ are disjoint. Therefore $\Pr^{G,P}_{sk,\sigma_{\text{AAF}}}(a) + \Pr^{G,P}_{sk,\sigma_{\text{AAF}}}(b) \leq \sum_{G_{\downarrow W} \in \Omega_{PrG}} \Pr(W) = 1$.

Thus every $PrG \in \mathscr{U}_{PAAF}$ satisfies the counterparts of Postulates 1–3. □

Postulate 4 (Subargument Closure)**.** *$PrF \in \mathscr{U}_{PSAF}$ satisfies* subargument closure *iff for every σ_{SAF} and every $E \subseteq A$, if $\bigcup_{a \in E} Sub^*(a) \neq E$, then $\Pr^{F,P}_{\sigma_{\text{SAF}}}(E) = 0$.*

Postulate 5 (Subargument Monotonicity)**.** *$PrF \in \mathscr{U}_{PSAF}$ satisfies* subargument monotonicity *iff for every σ_{SAF}, every $\circ \in \{cr, sk\}$ and every $a,b \in A$, if $a \in Sub^*(b)$, then $\Pr^{F,P}_{\circ,\sigma_{\text{SAF}}}(b) \leq \Pr^{F,P}_{\circ,\sigma_{\text{SAF}}}(a)$.*

Postulate 6 (Defence Bound)**.** *$PrF \in \mathscr{U}_{PSAF}$ satisfies* defence bound *iff for every $a \in A$, every admissible set $E \subseteq A$ and every $\sigma_{\text{SAF}} \in \{CMP, PRF\}$, if $a \in E$, then $\Pr^{F,P}_{cr,\sigma_{\text{SAF}}}(a) \geq \sum_{F_B \in \Omega_{PrF}, E \subseteq A_B} \Pr(F_B)$.*

Postulate 7 (Defence Sufficiency)**.** *$PrF \in \mathscr{U}_{PSAF}$ satisfies* defence sufficiency *iff for every $a \in A$ and every $\sigma_{\mathrm{SAF}} \in \{CMP, PRF\}$, if there exists an admissible set $E \subseteq A$ s.t. $a \in E$ and for every $b \in E$, $\mathrm{Sub_b}(b) \subseteq \mathrm{Sub_b}(a)$, then $\mathrm{Pr}^{F,P}_{cr,\sigma_{\mathrm{SAF}}}(a) = P(a)$.*

Theorem 4. *Every $PrF \in \mathscr{U}_{PSAF}$ satisfies Postulates 4–7.*

Proof. Let $PrF = (F,P)$, where $F = (A,D,Sub)$, and let Ω_{PrF} be the set of all possible worlds of PrF.

For Postulate 4, fix σ_{SAF} and $E \subseteq A$ such that $\bigcup_{a\in E} Sub^*(a) \neq E$. By definition, every σ_{SAF}-extension of every possible world is admissible, and hence closed under subarguments. Therefore E is not a σ_{SAF}-extension of any $F_B \in \Omega_{PrF}$. Hence $\mathrm{Pr}^{F,P}_{\sigma_{\mathrm{SAF}}}(E) = \sum_{F_B\in\Omega_{PrF}, E\in\sigma_{\mathrm{SAF}}(F_B)} \mathrm{Pr}(F_B) = 0$.

For Postulate 5, fix σ_{SAF}, $\circ \in \{cr, sk\}$ and $a,b \in A$ with $a \in Sub^*(b)$. Let $F_B \in \Omega_{PrF}$. If b is credulously accepted in F_B, then some σ_{SAF}-extension of F_B contains b; since every such extension is closed under subarguments, it also contains a, so a is credulously accepted in F_B. If b is skeptically accepted in F_B, then every σ_{SAF}-extension of F_B contains b, and therefore also a; thus a is skeptically accepted in F_B. Hence $\{F_B \in \Omega_{PrF} \mid F_B \mid\!\sim^{\circ}_{\sigma_{\mathrm{SAF}}} b\} \subseteq \{F_B \in \Omega_{PrF} \mid F_B \mid\!\sim^{\circ}_{\sigma_{\mathrm{SAF}}} a\}$, and so $\mathrm{Pr}^{F,P}_{\circ,\sigma_{\mathrm{SAF}}}(b) \leq \mathrm{Pr}^{F,P}_{\circ,\sigma_{\mathrm{SAF}}}(a)$.

For Postulate 6, fix $a \in A$, an admissible set $E \subseteq A$ with $a \in E$, and $\sigma_{\mathrm{SAF}} \in \{CMP, PRF\}$. Let $F_B \in \Omega_{PrF}$ with $E \subseteq A_B$. We show that E is admissible in $F_B = (A_B, D_B, Sub_B)$. Since $D_B \subseteq D$ and $Sub_B \subseteq Sub$, conflict-freeness and subargument closure of E are preserved in F_B. For defence, let $e \in E$, let $x \in Sub^*(e)$, and let $d \in A_B$ satisfy $(d,x) \in D_B$. Since E is admissible in F, there exist $c \in E$ and $y \in Sub^*(d)$ such that $(c,y) \in D$. As $d \in A_B$ and F_B satisfies subargument closure, we have $y \in A_B$, so $(c,y) \in D_B$. Hence E is admissible in F_B. Since every admissible set in a finite SAF is contained in some preferred extension, and every preferred extension is complete, a is credulously accepted in F_B under both PRF and CMP. Therefore $\{F_B \in \Omega_{PrF} \mid E \subseteq A_B\} \subseteq \{F_B \in \Omega_{PrF} \mid F_B \mid\!\sim^{cr}_{\sigma_{\mathrm{SAF}}} a\}$, and thus $\mathrm{Pr}^{F,P}_{cr,\sigma_{\mathrm{SAF}}}(a) \geq \sum_{F_B\in\Omega_{PrF}, E\subseteq A_B} \mathrm{Pr}(F_B)$.

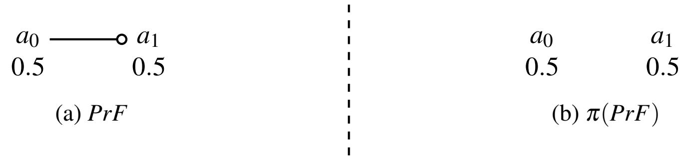
For Postulate 7, fix $a \in A$ and $\sigma_{\mathrm{SAF}} \in \{CMP, PRF\}$, and suppose there exists an admissible set $E \subseteq A$ such that $a \in E$ and $\mathrm{Sub_b}(b) \subseteq \mathrm{Sub_b}(a)$ for every $b \in E$. Let $F_B \in \Omega_{PrF}$ with $a \in A_B$. By Lemma 1, $\mathrm{Sub_b}(a) \subseteq \mathrm{Base}(F_B)$. Hence for every $b \in E$, $\mathrm{Sub_b}(b) \subseteq \mathrm{Sub_b}(a) \subseteq \mathrm{Base}(F_B)$, and again by Lemma 1, $b \in A_B$. Thus $E \subseteq A_B$ whenever $a \in A_B$. By Postulate 6, $\mathrm{Pr}^{F,P}_{cr,\sigma_{\mathrm{SAF}}}(a) \geq \sum_{F_B\in\Omega_{PrF}, a\in A_B} \mathrm{Pr}(F_B) = P(a)$, where the equality is Proposition 4. On the other hand, by Postulate 2, $\mathrm{Pr}^{F,P}_{cr,\sigma_{\mathrm{SAF}}}(a) \leq P(a)$. Therefore $\mathrm{Pr}^{F,P}_{cr,\sigma_{\mathrm{SAF}}}(a) = P(a)$.

Thus every $PrF \in \mathscr{U}_{PSAF}$ satisfies Postulates 4–7. □

Proposition 5 (Violation of Postulate 4)**.** *There exists a $PrF \in \mathscr{U}_{PSAF}$ and a set $E \subseteq A$ such that $\bigcup_{a\in E} Sub^*(a) \neq E$, but in the projected PAAF $\pi(PrF) = (\Gamma(F), P)$, $\mathrm{Pr}^{\Gamma(F),P}_{\sigma_{\mathrm{AAF}}}(E) > 0$ for every $\sigma_{\mathrm{AAF}} \in \{cmp, grd, prf, stb\}$.*

Proof. Consider the PSAF $PrF = (F,P)$, where $F = (A,D,Sub)$, $A = \{a_0, a_1\}$, $D = \varnothing$, and $Sub = \{(a_0, a_1)\}$. Let $P(a_0) = P(a_1) = 0.5$ and let $E = \{a_1\}$ as shown in Figure 1a. Then $Sub^*(a_1) = \{a_0, a_1\}$, so $\bigcup_{a\in E} Sub^*(a) = \{a_0, a_1\} \neq E$.

Now consider the projected PAAF $\pi(PrF) = (\Gamma(F), P)$. Since $D = \varnothing$, its forgetful projection is $\Gamma(F) = (A, \varnothing)$ as shown in Figure 1b. Hence every possible world of



(a) PrF (b) $\pi(PrF)$

Figure 1.

$\Gamma(F)$ is an attack-free AAF. In particular, let $W = \{a_1\}$. Since arguments are treated as probabilistically primitive in the projected PAAF, the probability of $\Gamma(F)_{\downarrow W}$ is

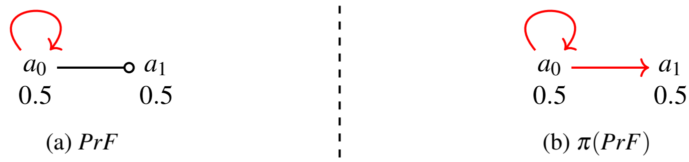
$$\Pr(\Gamma(F)_{\downarrow W}) = P(a_1)(1 - P(a_0)) = 0.25 > 0.$$

Because $\Gamma(F)_{\downarrow W}$ has argument set W and no attacks, $E = W$ is the unique complete extension of $\Gamma(F)_{\downarrow W}$, and therefore also its unique grounded, preferred, and stable extension. Thus, for every $\sigma_{\text{AAF}} \in \{cmp, grd, prf, stb\}$,

$$\Pr^{\Gamma(F),P}_{\sigma_{\text{AAF}}}(E) \geq \Pr(\Gamma(F)_{\downarrow W}) = 0.25 > 0.$$

Therefore there exists $PrF \in \mathscr{U}_{PSAF}$ and a set $E \subseteq A$ such that $\bigcup_{a \in E} Sub^*(a) \neq E$, but in the projected PAAF $\pi(PrF) = (\Gamma(F), P)$, for every corresponding classical semantics σ_{AAF}, $\Pr^{\Gamma(F),P}_{\sigma_{\text{AAF}}}(E) > 0$. □

Proposition 6 (Violation of Postulate 5)**.** *There exist a $PrF \in \mathscr{U}_{PSAF}$ and $a, b \in A$ such that $a \in Sub^*(b)$, but in the projected PAAF $\pi(PrF) = (\Gamma(F), P)$, $\Pr^{\Gamma(F),P}_{\circ,\sigma_{\text{AAF}}}(b) > \Pr^{\Gamma(F),P}_{\circ,\sigma_{\text{AAF}}}(a)$ for every acceptance mode $\circ \in \{cr, sk\}$ and every $\sigma_{\text{AAF}} \in \{cmp, grd, prf, stb\}$.*



(a) PrF (b) $\pi(PrF)$

Figure 2.

Proof. Consider the PSAF $PrF = (F, P)$, where $F = (A, D, Sub)$, $A = \{a_0, a_1\}$, $D = \{(a_0, a_0)\}$, and $Sub = \{(a_0, a_1)\}$. Thus $a_0 \in Sub^*(a_1)$. Let $P(a_1) = P(a_0) = 0.5$ as shown in Figure 2.

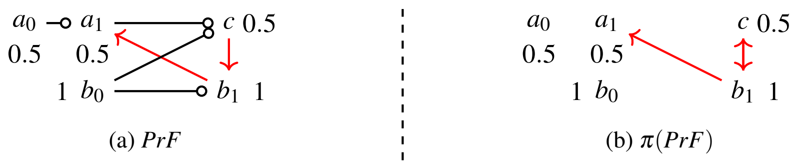
Now consider the projected PAAF $\pi(PrF) = (\Gamma(F), P)$. By Definition, since a_0 attacks itself and $a_0 \in Sub^*(a_1)$, the forgetful projection is $\Gamma(F) = (A, D_\Gamma)$ with $D_\Gamma = \{(a_0, a_0), (a_0, a_1)\}$. Hence the possible worlds of $\Gamma(F)$ are $\Gamma(F)_{\downarrow \emptyset}$, $\Gamma(F)_{\downarrow \{a_0\}}$, $\Gamma(F)_{\downarrow \{a_1\}}$, and $\Gamma(F)_{\downarrow \{a_0, a_1\}}$, each with probability 0.25.

We examine the four worlds. In $\Gamma(F)_{\downarrow \emptyset}$, no argument is accepted. In $\Gamma(F)_{\downarrow \{a_0\}}$, the only argument a_0 is self-attacking, so a_0 is neither credulously nor skeptically accepted under any of cmp, grd, prf, or stb. In $\Gamma(F)_{\downarrow \{a_1\}}$, the unique extension under each of cmp, grd, prf, and stb is $\{a_1\}$, so a_1 is both credulously and skeptically accepted. Finally, in $\Gamma(F)_{\downarrow \{a_0, a_1\}}$, the argument a_0 attacks itself and also attacks a_1, so neither a_0 nor a_1 is credulously or skeptically accepted under any of the four semantics. Therefore, for every acceptance mode $\circ \in \{cr, sk\}$ and every $\sigma_{\text{AAF}} \in \{cmp, grd, prf, stb\}$,

$$\Pr^{\Gamma(F),P}_{\circ,\sigma_{\text{AAF}}}(a_1) = \Pr(\{a_1\}) = 0.25 > \Pr^{\Gamma(F),P}_{\circ,\sigma_{\text{AAF}}}(a_0) = 0.$$

Thus there exist $PrF \in \mathscr{U}_{PSAF}$ and $a, b \in A$ such that $a \in Sub^*(b)$, but in the projected PAAF $\pi(PrF) = (\Gamma(F), P)$, $\Pr^{\Gamma(F),P}_{\circ,\sigma_{\text{AAF}}}(b) > \Pr^{\Gamma(F),P}_{\circ,\sigma_{\text{AAF}}}(a)$ for every $\circ \in \{cr, sk\}$ and every σ_{AAF}. $\square$

Proposition 7 (Violation of Postulate 6)**.** *There exist a $PrF \in \mathscr{U}_{PSAF}$, $a \in A$ and an admissible set $E \subseteq A$ such that $a \in E$, but in the projected PAAF $\pi(PrF) = (\Gamma(F), P)$, $\Pr^{\Gamma(F),P}_{cr,\sigma_{\text{AAF}}}(a) < \sum_{F_B \in \Omega_{PrF}, E \subseteq A_B} \Pr(F_B)$ for every $\sigma_{\text{AAF}} \in \{cmp, prf\}$.*



(a) PrF (b) $\pi(PrF)$

Figure 3.

Proof. Consider the PSAF $PrF = (F, P)$, where $F = (A, D, Sub)$, $A = \{a_0, a_1, b_0, b_1, c\}$, $D = \{(b_1, a_1), (c, b_1)\}$, and $Sub = \{(a_0, a_1), (b_0, b_1), (a_1, c), (b_0, c)\}$. Let $P(a_0) = P(a_1) = 0.5$, $P(b_0) = P(b_1) = 1$ and $P(c) = P(a_0)P(b_0) = 0.5$ as shown in Figure 3a.

Let $E = \{a_0, a_1, b_0, c\}$. Then $a_1 \in E$, E is closed under subarguments, and E is conflict-free. Moreover, the only attacker of a_1 is b_1, and it is attacked by c, so E is admissible in F.

Now consider the possible worlds of PrF. Since the basic arguments are a_0 and b_0, the condition $E \subseteq A_B$ holds exactly when $B = \{a_0, b_0\}$. Hence

$$\sum_{F_B \in \Omega_{PrF}, E \subseteq A_B} \Pr(F_B) = \Pr(F_{\{a_0, b_0\}}) = P(a_0)P(b_0) = 0.5.$$

Next consider the projected PAAF $\pi(PrF) = (\Gamma(F), P)$. By Definition, since b_1 attacks the subargument a_1 of c, the forgetful projection is $\Gamma(F) = (A, D_\Gamma)$ with $D_\Gamma = \{(b_1, a_1), (b_1, c), (c, b_1)\}$ as shown in Figure 3b. Thus b_1 is always present and attacks a_1. Therefore, in any possible world of $\Gamma(F)$, if c is absent, then a_1 is not credulously accepted under either cmp or prf. On the other hand, if both a_1 and c are present, then $\{a_1, c\}$ is a complete extension and also a preferred extension of the induced world. Hence, for each $\sigma_{\text{AAF}} \in \{cmp, prf\}$, a_1 is credulously accepted exactly in those worlds in which both a_1 and c are present. Since arguments are probabilistically primitive in the projected PAAF, this gives

$$\Pr^{\Gamma(F),P}_{cr,\sigma_{\text{AAF}}}(a_1) = P(a_1)P(c) = 0.25 < \sum_{F_B \in \Omega_{PrF}, E \subseteq A_B} \Pr(F_B) = 0.5$$

Thus there exist a $PrF \in \mathscr{U}_{PSAF}$, $a \in A$ and an admissible set $E \subseteq A$ such that $a \in E$, but in the projected PAAF $\pi(PrF) = (\Gamma(F), P)$, $\Pr^{\Gamma(F),P}_{cr,\sigma_{\text{AAF}}}(a) < \sum_{F_B \in \Omega_{PrF}, E \subseteq A_B} \Pr(F_B)$ for every $\sigma_{\text{AAF}} \in \{cmp, prf\}$. $\square$

Proposition 8 (Violation of Postulate 7)**.** *There exist a $PrF \in \mathscr{U}_{PSAF}$, $a \in A$ and an admissible set $E \subseteq A$ such that $a \in E$ and for every $b \in E$, $\mathrm{Sub_b}(b) \subseteq \mathrm{Sub_b}(a)$, but in the projected PAAF $\pi(PrF) = (\Gamma(F), P)$, $\mathrm{Pr}^{\Gamma(F),P}_{cr,\sigma_{\mathrm{AAF}}}(a) \neq P(a)$ for every $\sigma_{\mathrm{AAF}} \in \{cmp, prf\}$.*

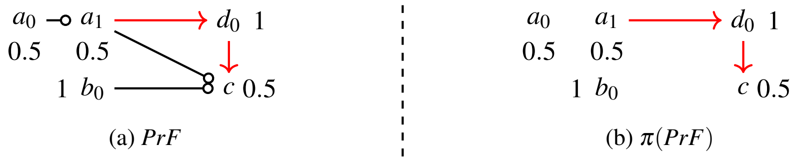


Figure 4.

Proof. Consider the PSAF $PrF = (F, P)$, where $F = (A, D, Sub)$, $A = \{a_0, a_1, b_0, c, d_0\}$, $D = \{(d_0, c), (a_1, d_0)\}$, and $Sub = \{(a_0, a_1), (a_1, c), (b_0, c)\}$. Let $P(a_0) = P(a_1) = 0.5$, $P(b_0) = 1$, $P(d_0) = 1$ and $P(c) = P(a_0)P(b_0) = 0.5$ as shown in Figure 4a.

Let $E = \{a_0, a_1, b_0, c\}$. Then $c \in E$. Moreover, E is closed under subarguments and conflict-free, and the only attacker of E is d_0, which is countered by a_1. Hence E is admissible in F. Also, $\mathrm{Sub_b}(a_0) = \{a_0\}, \mathrm{Sub_b}(a_1) = \{a_0\}, \mathrm{Sub_b}(b_0) = \{b_0\}, \mathrm{Sub_b}(c) = \{a_0, b_0\}$, so for every $b \in E$, $\mathrm{Sub_b}(b) \subseteq \mathrm{Sub_b}(c)$.

Now consider the projected PAAF $\pi(PrF) = (\Gamma(F), P)$. By definition, the forgetful projection is $\Gamma(F) = (A, D_\Gamma)$ with $D_\Gamma = \{(d_0, c), (a_1, d_0)\}$ as shown in Figure 4b.

Because $P(d_0) = 1$, the argument d_0 is present in every possible world of $\Gamma(F)$. Thus, in any possible world, if c is credulously accepted under either cmp or prf, then its attacker d_0 must be defeated, so a_1 must also be present. Conversely, whenever both a_1 and c are present, the set $\{a_1, c\}$ is an admissible set and extends to a complete extension and to a preferred extension of the induced world. Hence, for each $\sigma_{\mathrm{AAF}} \in \{cmp, prf\}$, c is credulously accepted exactly in those worlds in which both a_1 and c are present.

Since arguments are probabilistically primitive in the projected PAAF and retain the same marginals, for every $\sigma_{\mathrm{AAF}} \in \{cmp, prf\}$,

$$\mathrm{Pr}^{\Gamma(F),P}_{cr,\sigma_{\mathrm{AAF}}}(c) = P(a_1)P(c) = 0.25 \neq P(c) = 0.5$$

Thus there exist a $PrF \in \mathscr{U}_{PSAF}$, $a \in A$, and an admissible set $E \subseteq A$ such that $a \in E$ and for every $b \in E$, $\mathrm{Sub_b}(b) \subseteq \mathrm{Sub_b}(a)$, but in the projected PAAF $\pi(PrF) = (\Gamma(F), P)$, $\mathrm{Pr}^{\Gamma(F),P}_{cr,\sigma_{\mathrm{AAF}}}(a) \neq P(a)$ for every $\sigma_{\mathrm{AAF}} \in \{cmp, prf\}$. $\square$

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